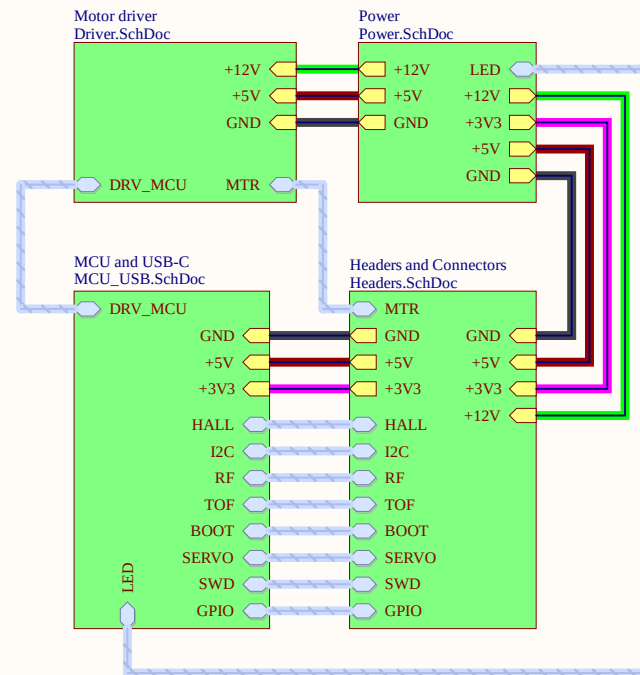
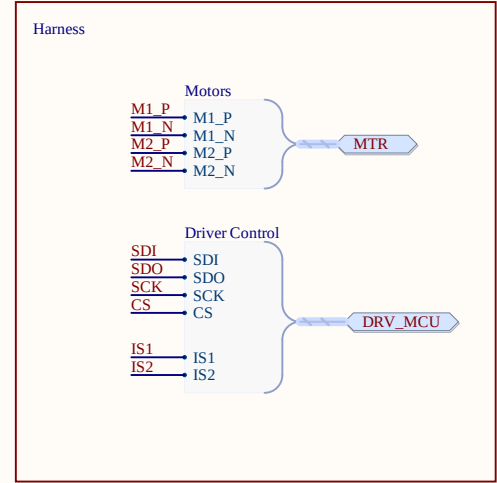
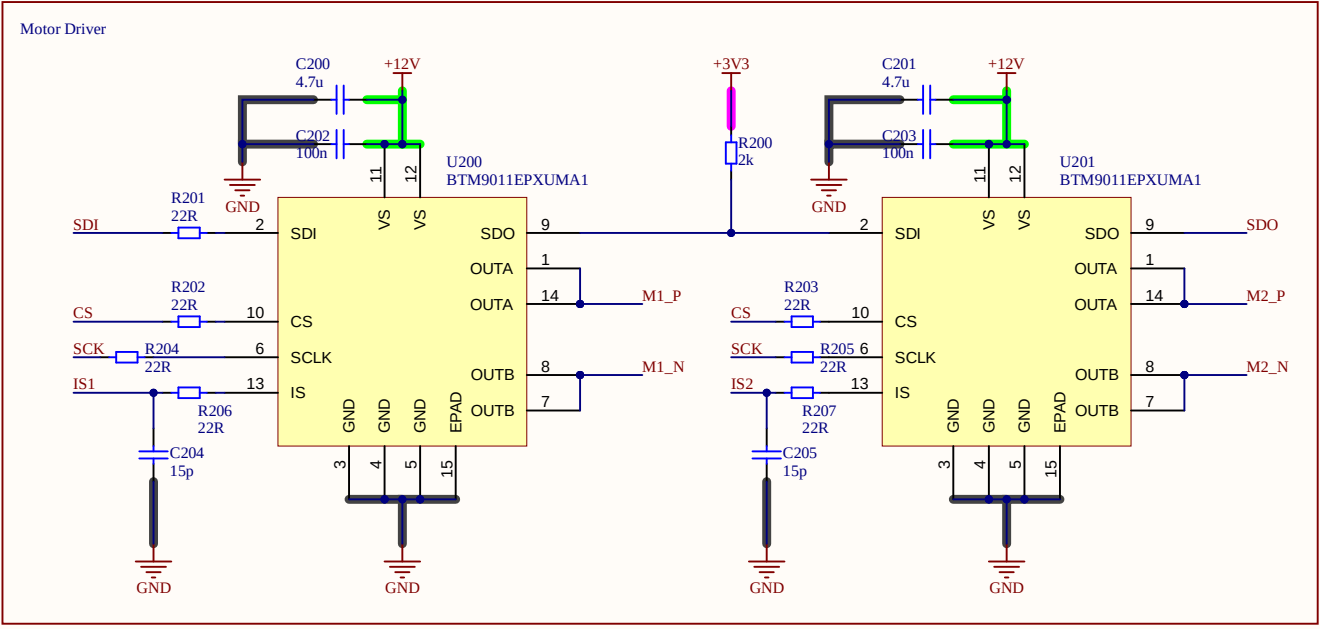
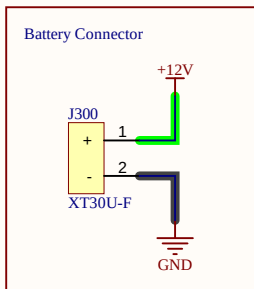
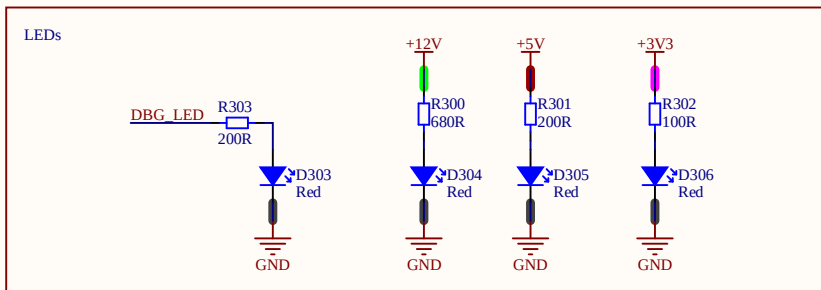
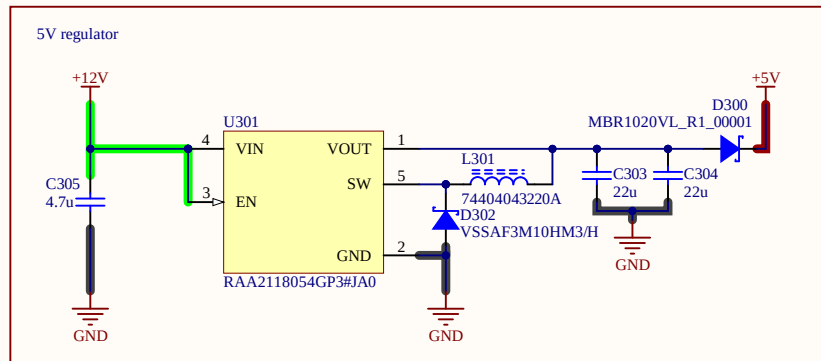
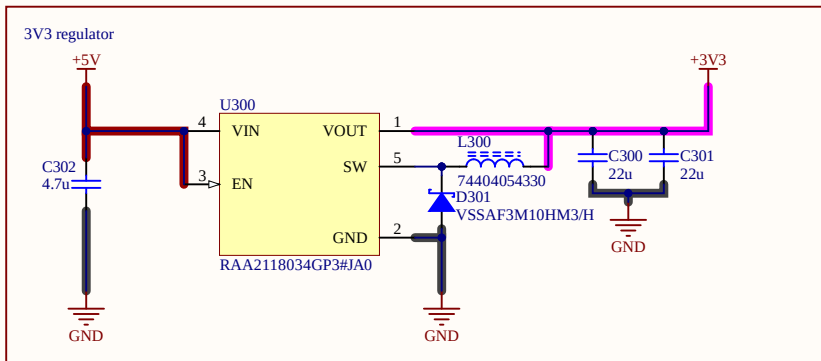
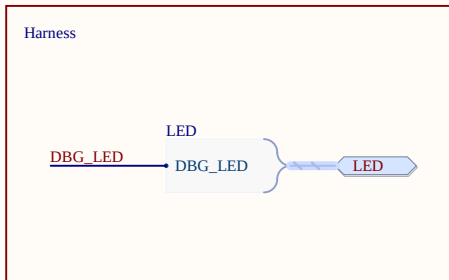




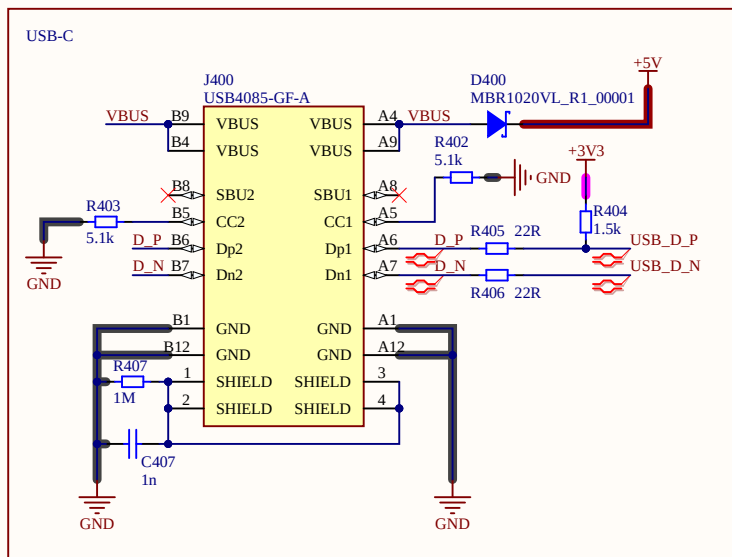
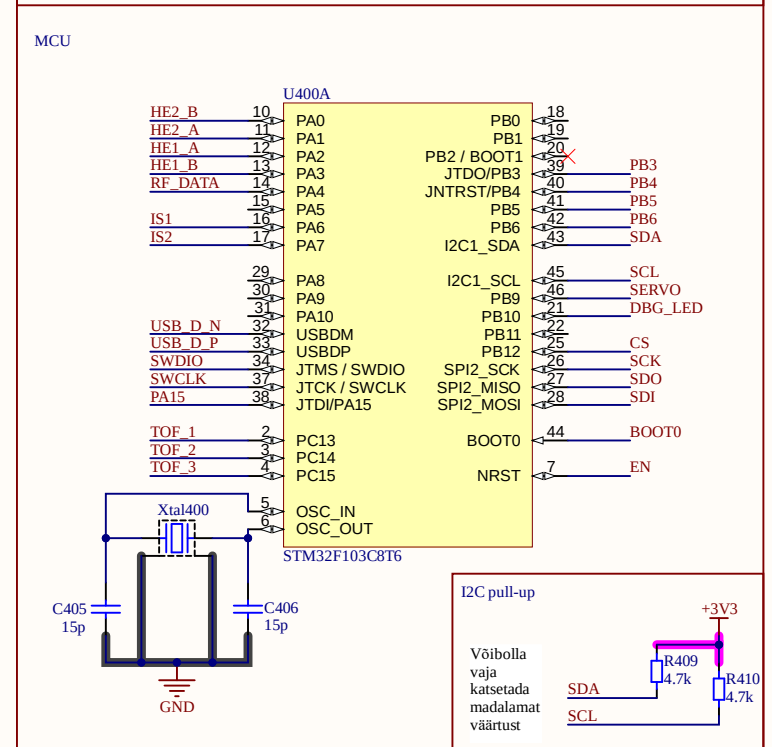
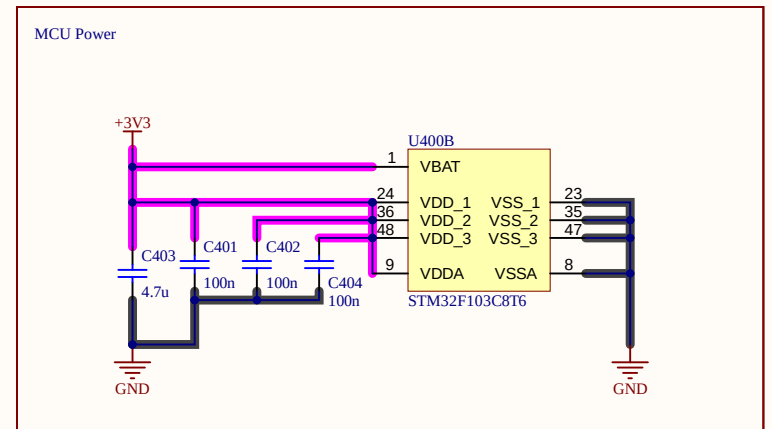
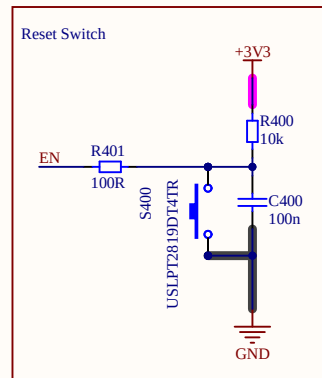
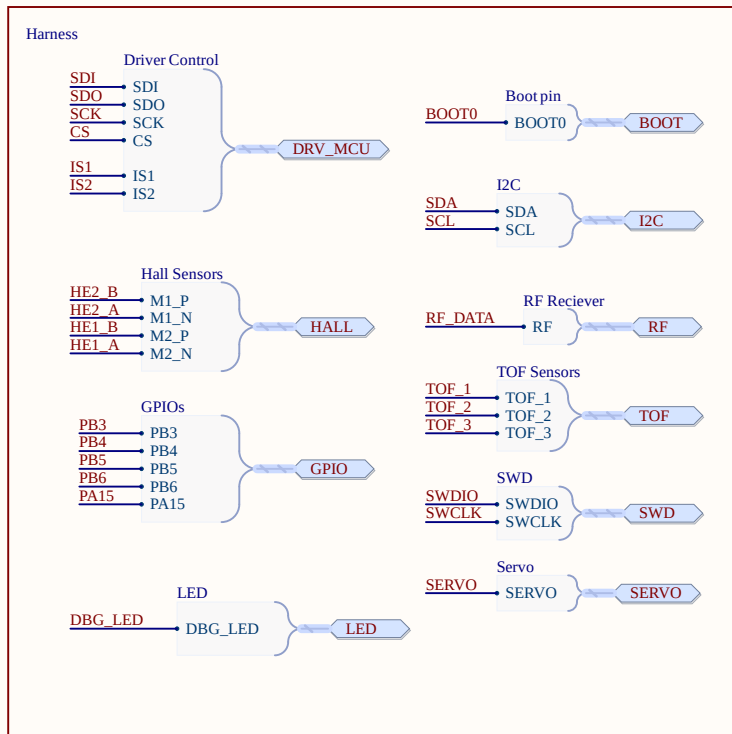
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Date:	1/31/2026	Sheet of
File:	C:\Users\...\Overview.SchDoc	Drawn By:
	5	6



Title		
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Date:	1/31/2026	Sheet of
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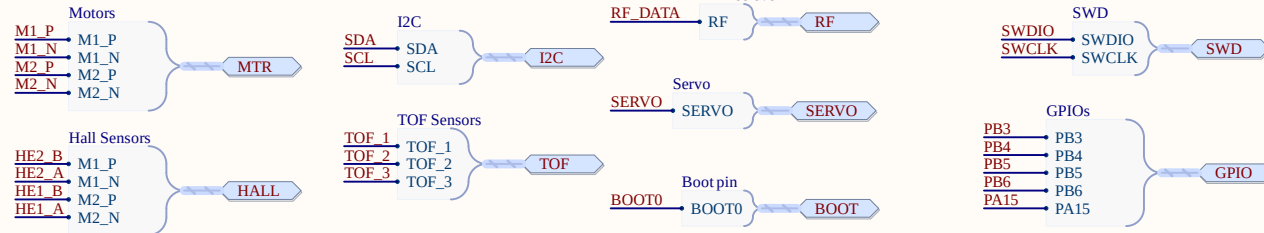


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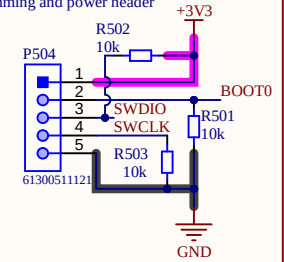


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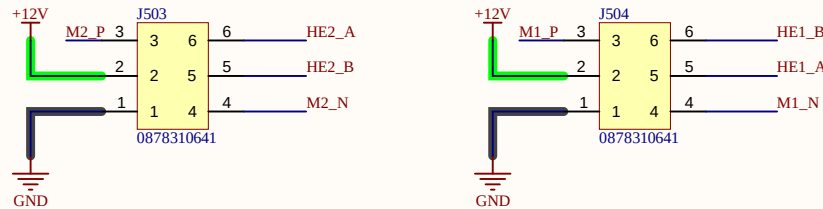
Harness



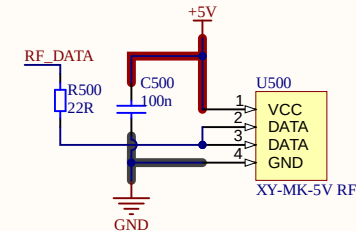
Boot, programming and power header



Motor Connectors

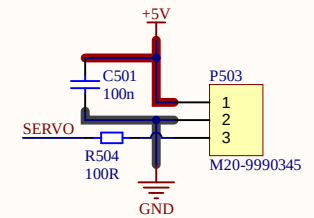


Receiver



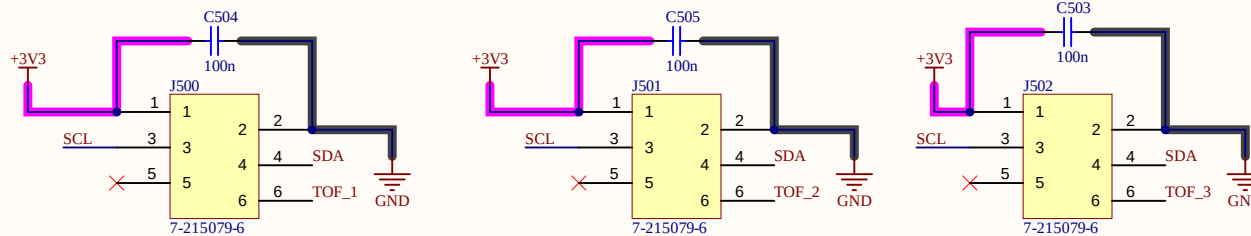
Receiver boards
solderable header

Servo



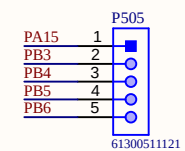
Servo motor
header

TOF Sensors



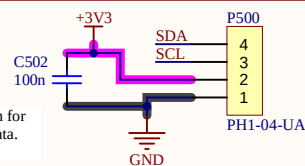
IDC connectors for TOF sensors.
2.54mm open headers connection
is unreliable

Extra pins



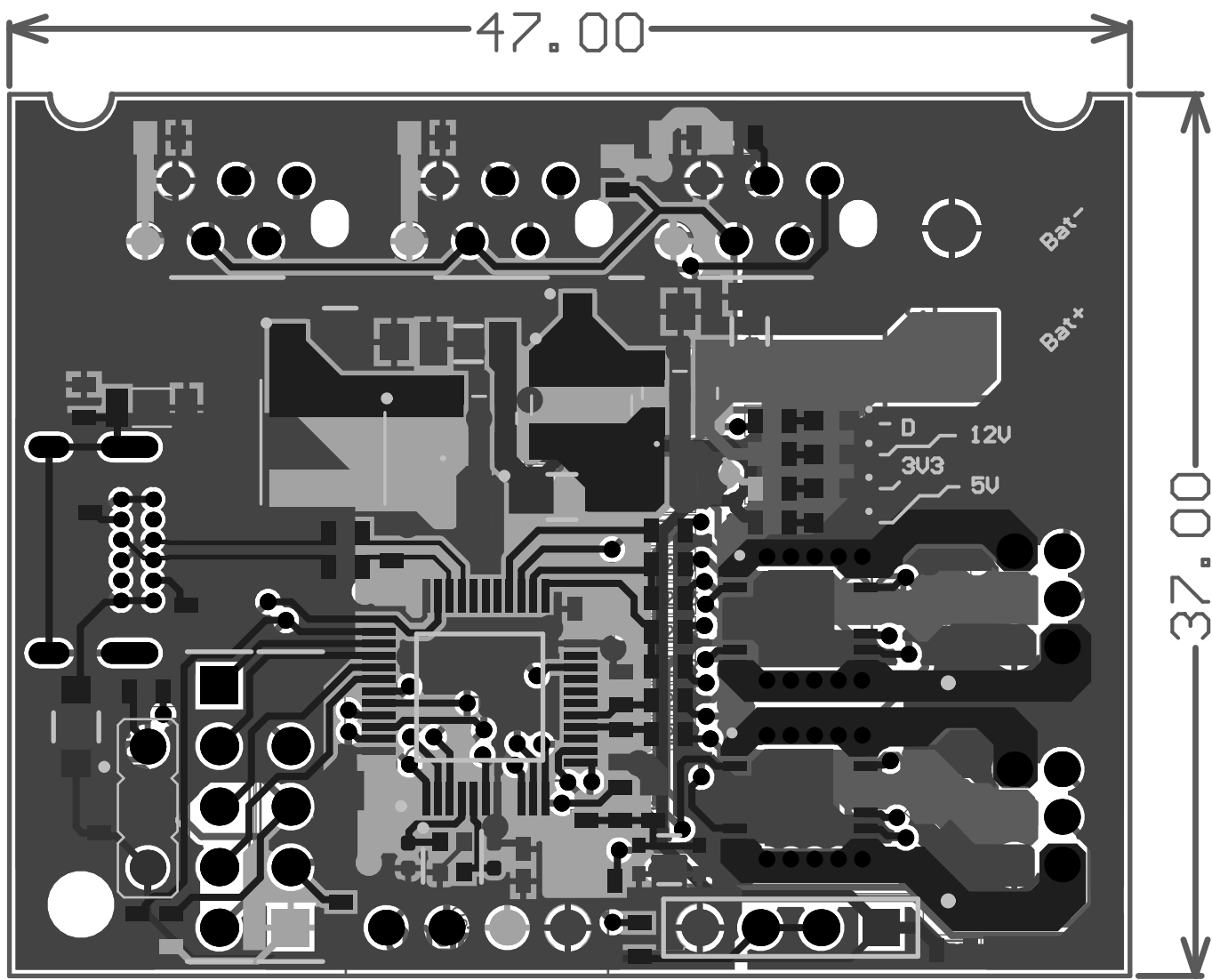
Extra header for
future use

OLED



OLED screen for
displaying data.

Title		
Size A3	Number	Revision
Date: 1/31/2026	Sheet of	
File: C:\Users\...\Headers.SchDoc	Drawn By:	



Comment	Description	Designator	Footprint	LibRef	Quantity
4.7u	MLCC, 0603, 4.7u, 10%, 35V, X5R	C200, C201	MLCC_0603_M	CMP-001-1257-1	2
100n	MLCC, 0603, 100n, 10%, 35V, X7R, MLCC, 0603, 100n, 10%, 50V, X7R	C202, C203, C400, C401, C402, C404, C500, C501, C502, C503, C504, C505	MLCC_0603_M	CMP-001-0755-1, CMP-001-0815-1	12
15p	MLCC, 0603, 15p, 5%, 50V, COG/NP0	C204, C205	MLCC_0603_M	CMP-001-0474-1	2
22u	MLCC, 0603, 22u, 10%, 35V, X5R	C300, C301, C303, C304	MLCC_0603_M	CMP-001-1264-1	4
4.7u	MLCC, 1206, 4.7u, 10%, 35V, X5R	C302, C305	MLCC_1206_M	CMP-001-1124-2	2
4.7u	MLCC, 0805, 4.7u, 20%, 50V, X5R	C403	MLCC_0805_M	CMP-001-1830-1	1
15p	MLCC, 0402, 15p, 10%, 25V, COG/NP0	C405, C406	MLCC_0402_M	CMP-001-1853-1	2
1n	MLCC, 0603, 1n, 10%, 50V, X5R	C407	MLCC_0603_M	CMP-001-0547-1	1
MBR1020M_R1_0000 1	Rectifier Diode, Schottky, 1 Phase, 1 Element, 1A, 20V V(RRM), Silicon	D300, D400	FP- MBR1020M_R1_0000 1-MFG	CMP-ROOT-01310-1	2
VSSAF3M10H-M/H	Rectifier Diode, Schottky, 1 Phase, 1 Element, 2.3A, 100V V(RRM), Silicon, DO- 221AC	D301, D302	VSSAF3M10H-M_H- Footprint-1	CMP-ROOT-01308-1	2
Red	Generic red 0603 LED	D303, D304, D305, D306	0603 RED LED	CMP-ROOT-00025-2	4
XT30J-F	Power Connector	J300	XT30J-F	CMP-ROOT-01262-2	1
USB085-GF-A	USB Connector, 16 Contact(s), Female, Right Angle, Solder Terminal, Locking Receptacle	J400	FP-USB085-GF-A- MFG	CMP-001-1800-1	1
7-215079-6	Conn FFC Connector F 6 POS 1.27mm Solder ST Thru-Hole Automotive Box/Cartron	J500, J501, J502	7-215079-6-Footprint- 1	CMP-ROOT-01311-2	3
0878310641	Conn Shrouded Header (2 Sides) HDR 6 POS 2mm Solder ST Top Entry Thru-Hole Milli-Grid Tube	J503, J504	0878310641	CMP-ROOT-01318-3	2
74404054330	Inductor Power Semi- Shielded Wirewound 33uH 20%, 100MHz 1.2A 0.18Ohm DCR 2020 Automotive TFR	L300	WE-LQS_5040	CMP-007-0001-1	1
74404043220A	General Purpose Inductor, 22uH, 20%, 1 Element, Nickel-zinc Ferrite-Core, SMD, 1616	L301	WE-LQS_4025	CMP-ROOT-01309-1	1
PH1-04-UA	4 POS 2.54mm Vertical THT Header	P500	4Pos Pin Header Vertical	CMP-002-000004-1	1
M20-9990345	Connector Header Through Hole 3 position 0.100" (2.54mm)	P503	3Pos Pin Header	CMP-ROOT-00217-1	1
61300511121	Conn Unshrouded Header HDR POS 2.54mm Solder ST Top Entry Thru-Hole Bag	P504, P505	61300511121	CMP-ROOT-01020-1	2
2k	Resistor, 0603, 2k, 1%, 1/10W	R600	Resistor_0603_M	CMP-001-0084-2	1
22R	Resistor, 0603, 22R, 1%, 1/10W	R601, R602, R603, R604, R605, R606, R607, R608, R609, R600	Resistor_0603_M	CMP-001-0038-3	10
680R	Resistor, 0603, 680R, 1%, 1/10W	R600	Resistor_0603_M	CMP-001-0074-3	1
200R	Resistor, 0603, 200R, 1%, 1/10W	R601, R603	Resistor_0603_M	CMP-001-0060-2	2
100R	Resistor, 0603, 100R, 5%, 1/10W, Resistor, 0603, 100R, 1%, 1/10W	R602, R401, R504	Resistor_0603_M	CMP-001-00262-1, CMP-001-0054-5	3
10k	Resistor, 0603, 10k, 1%, 1/10W, Resistor, 0603, 10k, 5%, 1/10W	R400, R501, R502, R603	Resistor_0603_M	CMP-001-0101-2, CMP-001-00257-1	4
5.1k	Resistor, 0603, 5.1k, 1%, 1/10W	R402, R403	Resistor_0603_M	CMP-001-0094-2	2
1.5k	Resistor, 0603, 1.5k, 1%, 1/10W	R404	Resistor_0603_M	CMP-001-0081-2	1
1M	Resistor, 1206, 1M, 1%, 1/4W	R407	Resistor_1206_M	CMP-001-00051-1	1
4.7k	Resistor, 0603, 4.7k, 1%, 1/10W	R409, R410	Resistor_0603_M	CMP-001-0083-2	2
USLPT2819DT4TR	Keypad Switch, SPST, 0.05A, 1.96N, Solder Terminal, Panel Mount	S400	USLPT2819DT4TR- Footprint-1	CMP-ROOT-01290-1	1
BTM0011EP9UMAJ	DC Mtr Drive, Brush, 150VDC, TSOPEP- 14 Pin Compliant, Yes	U200, U201	BTM0011EP9UMAJ	CMP-00022-3	2
RAA2118034G93U40	Switching Regulator, Current-mode, 0.3A, 170kHz Switching Freq Max, PDSOS	U300	P5_0648_MFG	CMP-ROOT-0017-1	1
RAA2118054G93U40	Switching Regulator, Current-mode, 0.3A, 230kHz Switching Freq Max, PDSOS	U301	P5_0648_MFG	CMP-00019-3	1
STM32F103C8T6	MCU 32-Bit STM32F1 ARM Cortex M3 RISC 64KB Flash 2.5V/3.3V	U400	STM-LQFP48_M	CMP-00012-2	1
XY-MK-5V-RF	48-Pin LQFP Tray 453MHz Receiver module, 5V tolerant	U500	2.54mm 4 POS XY-MK- 5V-RF	CMP-00015-1	1
ABM11-16.000MHZ-12- NIG-T	Crystal 16MHz ±10ppm (Ta) ±15ppm (Stability) 12pF FUND 150Ohm 4-Pin Ultra Mini-CSMD TFR	X4#400	FP-ABM11-MFG	CMP-ROOT-01254-3	1